

Report No: 2

Team: Autopilot

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1 Introduction

This report presents the progress achieved during the second month of our project, building upon the system architecture and manual control framework developed in the first month. The primary focus during this phase was the implementation of a **real-time perception and control pipeline**, with emphasis on **lane detection, vehicle control, and visualization** for autonomous navigation.

Key developments include a multiprocessing-based lane detection module using classical computer vision techniques, a PID-based motor control system, and a real-time dashboard for monitoring perception and control states. The system has been designed to operate both in a standalone multiprocessing framework and within a ROS-based architecture, ensuring modularity and scalability. This report details the implementation, current status, and challenges encountered during this phase.

2 Planned activities

The activities planned are as follows:

- Lane detection and tracking
- Vehicle control based on lane perception

Lane detection and tracking

Status: Straight lane detection(completed) curved lane detection(ongoing)

Implementation: Lane detection was implemented as a dedicated multiprocessing module to enable real-time performance and clean separation from other system components. Camera frames are received through an inter-process queue and processed using OpenCV. The lane detection system uses a midpoint-based method to understand where the vehicle is positioned within the lane. The steps followed are:

1. Detection of Lane Sides

The system first detects lane markings on both sides of the road. Lines with a left-leaning direction are treated as the left lane boundary, while right-leaning lines are treated as the right lane boundary.

2. Stabilizing Lane Positions

Since multiple line segments may be detected, the system averages them to get a single, stable position for the left lane and the right lane. This helps reduce noise and false detections.

3. Finding the Lane Centre

The centre of the lane is calculated by taking the midpoint between the left and right lane positions. This midpoint represents the ideal position where the vehicle should be aligned.

4. Calculating Vehicle Deviation

The system compares the lane centre with the centre of the camera image to find how much the vehicle is shifted to the left or right.

5. Estimating Reliability

A confidence value is assigned based on how clearly the lane markings are detected. When both lane sides are visible, the confidence is high, leading to more reliable control decisions.

Detected lines are classified into left and right lanes based on slope thresholds. From these, the lane centre is computed relative to the image centre, and an average lane angle is estimated from detected slopes. A confidence metric is derived from the number of valid lane detections.

The module outputs structured lane information—including lane centre, angle, confidence, and error—to the control process. In addition, an annotated video frame with visual overlays (lane lines, centre line, ROI, and textual metrics) is generated for debugging and visualization.

3 Vehicle Control

Status: Completed

Implementation:

While the algorithm performs reliably on straight lanes, curved lane detection remains challenging due to incomplete or fragmented line detection. Variations in lighting conditions, shadows, and reflections affect edge extraction and Hough detection consistency. Ongoing work focuses on improving robustness through parameter tuning and adaptive preprocessing techniques.

Difficulties:

Integrating real-time video footage for object detection was a significant challenge due to its complexity. However, this was resolved, and the system now operates seamlessly in real-time, ensuring accurate detection and response.

The traffic sign detection model has been completed and fulfills all BFMC specifications, identifying all required signs and adapting its behavior dynamically.

4 General status of the project

The project has progressed from manual control to a perception-driven autonomous framework. Lane detection for straight paths has been successfully implemented and integrated with vehicle control. The control system responds dynamically to perception inputs, and real-time visualization provides comprehensive system insight.

Current development efforts are focused on improving curved lane detection reliability and enhancing robustness under varying lighting and environmental conditions. The modular multiprocessing and ROS-compatible architecture positions the system well for future extensions.